IEOR 221

Homework 6

Module 7 Options – BSM Equations (October 13, 2025)

1) A company's cash position, measured in millions of dollars, follows a generalized Wiener process with a drift rate of 0.5 per quarter and a variance rate of 4.0 per quarter. How high does the company's initial cash position have to be for the company to have a less than 5% chance of a negative cash position by the end of 1 year? (Hull 14.12)

2) The process for the stock price in the lecture is

$$\Delta S = \mu S \Delta t + \sigma S \Delta W_t \qquad (*)$$

where m and s are constant. Explain carefully the difference between this model and each of the following:

$$\Delta S = \mu \Delta t + \sigma \Delta W_t$$
$$\Delta S = \mu S \Delta t + \sigma \Delta W_t$$
$$\Delta S = \mu \Delta t + \sigma S \Delta W_t$$

Why is the model in equation (*) a more appropriate model of stock price behavior than any of these three alternatives? (Hull 14.17)

3) Suppose that a stock price S follows geometric Brownian motion with expected return μ and volatility σ:

$$dS = \mu S dt + \sigma S dW_t$$

What is the process followed by the variable S^n? Show that S^n also follows geometric Brownian motion. (Hull 14.19)

4) Suppose that a stock price has an expected return of 16% per annum and a volatility of 30% per annum. When the stock price at the end of a certain day is \$50, calculate the following:
(a) The expected stock price at the end of the next day
(b) The standard deviation of the stock price at the end of the next day
(c) The 95% confidence limits for the stock price at the end of the next day.
(Hull 14.22)
(Assume the stock price follows geometric Brownian motion, and there are 250 trading days per year, hence 1 day = 1/250 year)